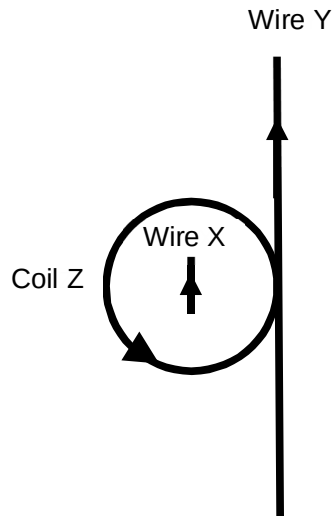


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The diagram below shows a setup of two current-carrying wires X and Y, and a single-turn coil Z.



Wire X has length 0.050 m with a current of 1.0 A flowing through it. Wire Y has length 2.0 m with a current of 0.50 A. Coil Z has a diameter of 0.20 m with a current of 1.5 A.

What is the resultant force on wire X if the centre of coil Z is on wire X?

A	$5.0 \times 10^{-8} \text{ N}$
B	$4.2 \times 10^{-7} \text{ N}$
C	$5.2 \times 10^{-7} \text{ N}$
D	$9.4 \times 10^{-6} \text{ N}$

Answer: C

By right-hand grip rule, the magnetic flux densities due to wire Y and due to coil Z are **both** out of the page.

Hence, the resultant magnetic flux density experienced by wire X

$$= \frac{\mu_0 NI}{2r} + \frac{\mu_0 I}{2\pi d} = \frac{4\pi \times 10^{-7} \times 1 \times 1.5}{2(0.1)} + \frac{4\pi \times 10^{-7} \times 0.5}{2\pi(0.1)}$$

$$= 1.042 \times 10^{-5} \text{ T}$$

Therefore, the resultant force on wire X = $BIL = 1.042 \times 10^{-5} \times 1.0 \times 0.05 = 5.2 \times 10^{-7} \text{ N}$